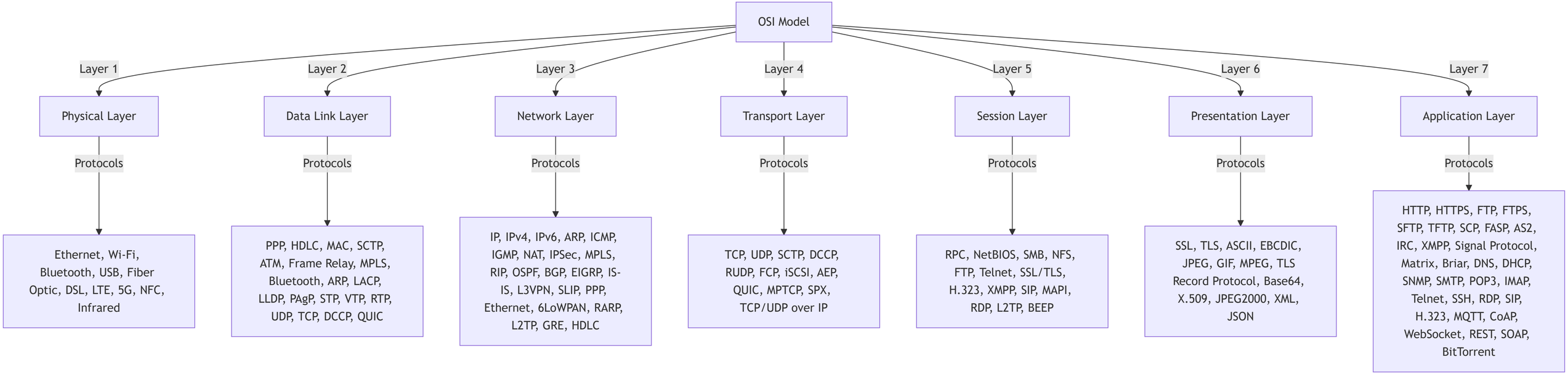
OSI Model
Layer 1
Physical Layer
Protocols
Ethernet, Wi-Fi, Bluetooth, USB, Fiber Optic, DSL, LTE, 5G, NFC, Infrared
Layer 2
Data Link Layer
Protocols
PPP, HDLC, MAC, SCTP, ATM, Frame Relay, MPLS, Bluetooth, ARP, LACP, LLDP, PAgP, STP, VTP, RTP, UDP, TCP, DCCP, QUIC
Layer 3
Network Layer
Protocols
IP, IPv4, IPv6, ARP, ICMP, IGMP, NAT, IPSec, MPLS, RIP, OSPF, BGP, EIGRP, IS-IS, L3VPN, SLIP, PPP, Ethernet, 6LoWPAN, RARP, L2TP, GRE, HDLC
Layer 4
Transport Layer
Protocols
TCP, UDP, SCTP, DCCP, RUDP, FCP, iSCSI, AEP, QUIC, MPTCP, SPX, TCP/UDP over IP
Layer 5
Session Layer
Protocols
RPC, NetBIOS, SMB, NFS, FTP, Telnet, SSL/TLS, H.323, XMPP, SIP, MAPI, RDP, L2TP, BEEP
Layer 6
Presentation Layer
Protocols
SSL, TLS, ASCII, EBCDIC, JPEG, GIF, MPEG, TLS Record Protocol, Base64, X.509, JPEG2000, XML, JSON
Layer 7
Application Layer
Protocols
HTTP, HTTPS, FTP, FTPS, SFTP, TFTP, SCP, FASP, AS2, IRC, XMPP, Signal Protocol, Matrix, Briar, DNS, DHCP, SNMP, SMTP, POP3, IMAP, Telnet, SSH, RDP, SIP, H.323, MQTT, CoAP, WebSocket, REST, SOAP, BitTorrent